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Discussion questions

1 Do you think that you have ever seen or used hexadecimal before? If you have, can you provide an example of where you have seen or used it?

2 Why is it important for a programmer to be able to effectively debug a computer program?

1.1 Binary and hexadecimal

Humans process a large amount of data on a daily basis. Did you know that when we have a conversation with someone, we process approximately 60 bits of information per second? Our brain is designed to process a maximum of about 120 bits per second. That's why it is so difficult to listen to two people talk to us at once!

As humans, we process data that is called **analogue data**. Everything we see, hear, feel and smell is a continuous stream of analogue data to our senses.

Questions

1 Can you think of **three** examples of analogue data?

2 Why is analogue data described as continuous data?

The components that process data in a computer are made up of very small **logic gates**. These gates cannot process analogue data, they can only process **digital data**. This data is stored in **registers**. You can find out more about logic gates in Chapter 10. Digital data is discrete data that only has two values, 1 and 0. Any data that we want a computer to process must be converted into digital data. This type of data is also called binary data.

KEY WORDS

analogue data: a continuous stream of data that is processed by humans.

logic gate: a very small component in a computer system that controls the flow of electricity.

digital data: discrete data that only uses the values 0 and 1.

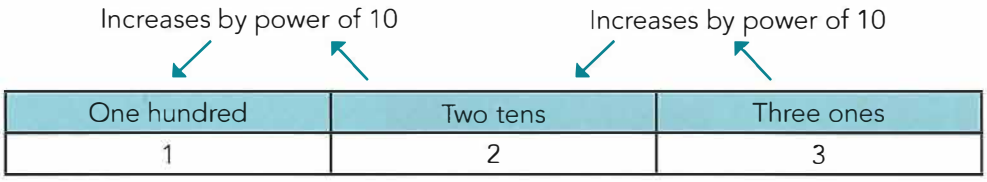
register: a small component in a computer system that is used to temporarily store data.

denary: a base-10 number system that uses the values 0–9.

Number systems

In our daily lives we mostly use a **denary** number system. This is called a base-10 number system and uses the digits 0–9. The units in a denary number system are ones, tens, hundreds and thousands. Each time the unit increases by the power of 10.

For example, the denary number 123 has:



One hundred	Two tens	Three ones
1	2	3

Computers use a binary number system. This is called a base-2 number system and uses the digits 0 and 1. The units in a binary number system increase by the power of 2.

Any denary values need to be converted to binary values so that they can be processed by a computer.

SKILLS FOCUS 1.1

CONVERTING BETWEEN DENARY AND BINARY NUMBERS

The size of a binary number is referred to by the number of bits it has. A 4-bit binary number will have 4 values that are either 1 or 0, for example, 1001. An 8-bit binary number will have 8 values that are also either 1 or 0. Binary values can get very large, some have as many as 64 bits; now that's a very big number! You will need to be able to understand binary numbers that have as many as 16 bits.

First, focus on converting denary numbers to 4-bit binary. To convert denary numbers to 4-bit binary the units that you need to use are 1, 2, 4 and 8. You can indicate with a 1 or a 0 whether that unit is required. For example, if you convert the denary number 13 to a 4-bit binary number you can use the following steps:

Write down the binary units from 8 to 1. An easy way to do this is using a table.

8	4	2	1

Do you notice how the units increase by the power of 2 each time (going from right to left)?

Compare the denary number (13) to the first binary unit (8) to see whether it is equal to or greater than the denary number. 13 is greater than 8, which means that the unit 8 is required and you can write a 1 beneath it. 8 can then be subtracted from the denary value, $13-8=5$.

8	4	2	1
1			

Now compare the denary number (5) to the next binary unit (4) to see whether it is equal to or greater than the denary number. 5 is greater than 4, which means that the unit 4 is required and you can write a 1 beneath it. 4 can then be subtracted from the denary value, $5-4=1$.

8	4	2	1
1	1		

Then compare the denary number (1) to the next binary unit (2) to see whether it is equal to or greater than the denary number. 1 is not greater than 2, which means that the unit 2 is not required and you can write a 0 beneath it. No subtraction is needed this time as you haven't used the unit.

8	4	2	1
1	1	0	

Finally, compare the denary number (1) to the last binary unit (1) to see whether it is equal to or greater than the denary number. 1 is equal to 1, which means that the unit is required and you can write a 1 beneath it. 1 can then be subtracted from the denary value. $1-1=0$.

8	4	2	1
1	1	0	1

Hexadecimal

Hexadecimal is also a number system. Humans can use hexadecimal, but computers cannot process it. Like denary numbers, hexadecimal numbers also need to be converted to binary to be processed by a computer.

Hexadecimal is a base-16 system and uses 16 symbols, these are the numbers 0–9 and the letters A–F. This is to allow the denary values 0–15 to be represented in hexadecimal. This is shown in Table 1.1.

Denary	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Table 1.1: Hexadecimal representations of denary numbers

The reason for using the symbols A–F for the denary numbers 10–15 is so only one symbol is needed for the denary number in hexadecimal.

SKILLS FOCUS 1.2

CONVERTING BETWEEN DENARY NUMBERS AND HEXADECIMAL

To convert a denary number to hexadecimal, it is easier to convert it to binary first. For example:

First convert the denary number 7 to 4-bit binary.

8	4	2	1
0	1	1	1

To convert the binary number to hexadecimal you can use simple addition again. 4 + 2 + 1 = 7. If you look at the symbol table (Table 1.1), you can see that the hexadecimal for 7 is also 7. Therefore, the denary number 7 is 0111 as binary and 7 as hexadecimal.

So, what happens with denary numbers that are greater than 9?

First convert the denary number 12 to 4-bit binary.

8	4	2	1
1	1	0	0

To convert the binary number to hexadecimal you can use simple addition again. 8 + 4 = 12. If you look at the symbol table (Table 1.1), you can see that the hexadecimal for 12 is C. Therefore, the denary number 12 is 1100 as binary and 12 as hexadecimal.

Converting the denary number to binary first isn't necessary for denary numbers that are less than 16, but it makes the process easier when you need to convert larger denary numbers.

So, how do you convert a larger denary number to hexadecimal?